

Ratios & Simplest Form — Answers

Final answers with a one-line reason. For full methods, see the worked examples in the notes.

Objective

Q1 (b) **4 : 3**. Divide 24 : 18 by 6.

Q2 (a) **60 : 40**. $60 : 40 = 3 : 2$.

Q3 **2 : 3**. Divide 600 : 900 by 300.

Q4 (b) **5 : 4**. 1 kg = 1000 g, so $1000 : 800 = 5 : 4$.

Q5 **False**. Order matters in a ratio.

Q6 (a) **Both true, R explains A**. Equal simplest forms $\Rightarrow$ proportional.

Short answer

Q7 (i) **3 : 4** (ii) **5 : 2**. (Divide by 15 and by 7.)

Q8 **Yes, proportional**. $21 : 6 = 7 : 2$ and $35 : 10 = 7 : 2$ (also $21 \times 10 = 6 \times 35 = 210$).

Q9 **3 : 5**. Divide 6 : 10 by 2.

Long answer

Q10 **A and C are similar**. $60 : 40 = 3 : 2$ and $30 : 20 = 3 : 2$; E is $60 : 60 = 1 : 1$, so it is different.

Q11 **1 : 10, then 4 : 13 — not the same**. $3 : 30 = 1 : 10$; $12 : 39 = 4 : 13$. Adding the same number to both terms changes the ratio.

Q12 **4 : 1**. 2 m = 200 cm, so $200 : 50 = 4 : 1$.

Total: 21 marks.